

DS701 Homework 1: Question Types; Vectors, Norms, Dot Products

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Name: _____ BU ID: _____ Section: A1 / B1

How this works. Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it without a computer or an AI assistant. It is **participation-graded**: a serious attempt at every problem earns full credit, and worked solutions are released the day after the deadline. Budget 1–2 hours.

Covers: Lecture 01 (What is Data Science? — the question taxonomy) and the linear-algebra essentials of Lecture 02 (vectors, norms, dot products; refreshed in A7). Note the Wednesday deadline: it falls after the Monday Sep 14 session on distances, so you will have seen the Lecture 02 material before this is due. **Reference sheet:** the Quiz 1 reference sheet publishes Wed Sep 30, after this set is due, so everything you need here is stated in the problem. From Sep 30 study with it — sections 01 and 02 cover this material, and it is what you will have on quiz day. This is a warm-up week: the arithmetic is small on purpose; the point is to get the definitions into your hands.

Problem 1: Classify the question

Practices: naming the question type before choosing a technique (Lecture 01, sheet §01).

A music-streaming service is planning its analytics for the fall. Six questions come up in the planning meeting:

1. What was the median listening time per user last month, broken down by country?
 2. Do our users fall into a few natural groups by the genres they listen to — something worth a closer look?
 3. Across our whole user base (not just the 2,000 users we surveyed), do people who listen during a commute play more podcasts than people who do not?
 4. Will *this* user cancel their subscription in the next 30 days?
 5. If we give every free-trial user a personalized playlist on day 1, will more of them convert to paid?
 6. By exactly which route does a personalized playlist raise conversion — does it lengthen sessions, which builds a daily habit, which then drives payment?
- (a) Classify each of the six questions using the lecture’s taxonomy (descriptive, exploratory, inferential, predictive, causal, mechanistic). For each, give **one line** saying what data and

what kind of analysis would answer it (a summary table, an experiment, a model scored on held-out users, ...). Name the word or phrase in the question that decided your classification.

#	type	deciding word/phrase	data & analysis (one line)
1			
2			
3			
4			
5			
6			

- (b) Question 4 is predictive. Rewrite it as a **causal** question about the same subscription decision — one sentence. Then say, in two or three sentences, what changes: what counts as evidence for the predictive version, what counts as evidence for the causal version, and why a very accurate churn model does not by itself answer the causal one.

Causal rewrite: _____

- (c) A colleague proposes answering question 5 by comparing conversion rates of past trial users who *happened* to build their own playlist on day 1 against those who did not, and reports “playlist users converted at twice the rate”. Which question type does this comparison actually answer? Name one plausible reason the two groups differ that has nothing to do with the playlist, and say what design would answer question 5 as asked. Keep it to a short paragraph.

The comparison actually answers a(n) _____ question.

Problem 2: Two vectors by hand

Practices: norms, dot product, cosine similarity, unit vectors (sheet §02; A7).

Let $\mathbf{x} = (1, 2, 2)$ and $\mathbf{y} = (2, 2, 1)$ in $\mathbb{R}^3$. Definitions you need: $\|\mathbf{v}\|_1 = \sum_i |v_i|$, $\|\mathbf{v}\|_2 = \sqrt{\sum_i v_i^2}$, $\|\mathbf{v}\|_\infty = \max_i |v_i|$, $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = \sum_i x_i y_i$, and $\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}$. A *unit vector* has ℓ_2 norm 1.

- (a) Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$, $\|\mathbf{x}\|_\infty$ and the same three norms of $\mathbf{y}$.

	ℓ_1	ℓ_2	ℓ_∞
$\mathbf{x}$	_____	_____	_____
$\mathbf{y}$	_____	_____	_____

- (b) Compute $\mathbf{x} \cdot \mathbf{y}$, the cosine similarity $\cos(\mathbf{x}, \mathbf{y})$ (as an exact fraction), and the angle between them (leave it as arccos of your fraction, and give degrees if you have a calculator). Then write down the unit vector in the direction of $\mathbf{x}$ and verify by direct computation that its ℓ_2 norm is 1.

$$\mathbf{x} \cdot \mathbf{y} = \text{_____} \quad \cos(\mathbf{x}, \mathbf{y}) = \text{_____} \quad \theta = \text{_____}$$

$$\text{Unit vector } \mathbf{x}/\|\mathbf{x}\|_2 = \text{_____}$$

- (c) Show that for **every** $\mathbf{v} \in \mathbb{R}^n$,

$$\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1.$$

A short argument is expected (three or four lines): for the left inequality, compare $\max_i v_i^2$ with $\sum_i v_i^2$; for the right, expand $(\sum_i |v_i|)^2$. Check the chain on your numbers from (a).

- (d) Give a nonzero vector for which all three norms are **equal**, and say in one line what has to be true of a vector for that to happen.

$\mathbf{v} =$ _____

Problem 3: Data as a matrix

Practices: rows = objects, columns = features; centering; the dot product of two feature columns as a first measure of “moving together” (sheet §02).

Four students report hours studied and their exam score:

student	hours (h)	score (s)
1	2	60
2	4	70
3	6	70
4	8	80

Written as a data matrix X this is 4×2 : $m = 4$ objects (rows), $n = 2$ features (columns). To **mean-center** a column, subtract that column's mean from each of its entries.

- (a) Compute the mean of each column and write out the centered matrix X_c (the two centered columns $\mathbf{h}_c$ and $\mathbf{s}_c$). Verify that each centered column sums to zero, and explain in one line why that must always happen.

$$\bar{h} = \underline{\hspace{2cm}} \quad \bar{s} = \underline{\hspace{2cm}}$$

- (b) Compute the dot product $\mathbf{h}_c \cdot \mathbf{s}_c$ of the two centered columns. What does its **sign** tell you about how the two features move together across students? (Think about which students contribute positive terms and which negative.)

$$\mathbf{h}_c \cdot \mathbf{s}_c = \underline{\hspace{2cm}}$$

- (c) Compute the cosine similarity of $\mathbf{h}_c$ and $\mathbf{s}_c$ (three decimals; $\sqrt{4000} \approx 63.25$). What would a value near 1, near 0, or near -1 say about hours and scores?

$\cos(\mathbf{h}_c, \mathbf{s}_c) = \underline{\hspace{2cm}}$

- (d) Suppose the scores had been recorded as fractions (0.60, 0.70, 0.70, 0.80) instead of percentages. Without redoing everything, say what happens to the dot product in (b) and to the cosine in (c), and one sentence on why that makes the cosine the more useful of the two as a summary of “how related”.

New dot product: $\underline{\hspace{2cm}}$ New cosine: $\underline{\hspace{2cm}}$